

Notes: Handel (AER, 2013)

Adverse Selection and Inertia in Health Insurance Markets: When Nudging Hurts

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1 Big picture

- **Question.** What happens when policymakers reduce consumer inertia in a health insurance market where adverse selection is already present?
- **Setting.** The paper studies employees at a large firm that redesigned its insurance menu, forced workers to actively choose plans in the redesign year, and then restored passive default renewal in later years.
- **Key empirical advantage.** The same families are observed in an active choice year and in later passive choice years with the same PPO option set, allowing inertia to be separated from stable preference heterogeneity.
- **Main descriptive fact.** Many employees keep dominated or clearly worse plans even after large premium changes, implying substantial inertia.
- **Structural model.** The paper estimates a health plan choice model with heterogeneous inertia, heterogeneous risk aversion, plan differentiation, and individual-level ex ante health risk distributions.
- **Main inertia estimate.** In the primary specification, the average implied value of inertia is about \$2,032 per family per year, a very large number relative to annual medical spending and premium differences.
- **Adverse selection fact.** Sicker and more expensive employees select into more comprehensive plans. This means that when healthy consumers leave generous coverage, premiums for that coverage rise.
- **Counterfactual result.** Reducing inertia improves individual choices conditional on prices, but the resulting re-sorting worsens adverse selection and can reduce aggregate welfare once premiums reprice endogenously.
- **“When nudging hurts.”** A policy that moves people toward their individually preferred plans can make the market equilibrium worse because it changes the risk composition and prices of plans.
- **Economic takeaway.** Behavioral frictions and adverse selection are not independent market failures. Removing one friction can amplify the other.

2 Incremental contribution of the paper

- **Clean inertia identification.** Earlier empirical work on switching costs often struggles to distinguish true inertia from persistent unobserved tastes. This paper uses a forced active re-enrollment event followed by passive default renewal to identify inertia separately from stable plan preferences.
- **Behavioral IO plus insurance selection.** The paper connects behavioral choice frictions to the equilibrium economics of health insurance. It does not stop at documenting poor plan choice; it embeds inertia into an adverse selection environment with endogenous premiums.
- **Quantification in dollar terms.** Inertia is estimated as an implied monetary cost of moving away from the incumbent plan. This makes the magnitude comparable to premiums, out-of-pocket spending, and welfare effects.
- **Link between nudges and equilibrium prices.** The counterfactual exercises show that improving individual decision-making can worsen equilibrium prices by sharpening selection across plans.
- **Distributional analysis.** The paper shows that switchers gain from reduced inertia, but non-switchers, low-income employees, and some high-coverage enrollees can lose because premiums adjust.
- **Policy contribution.** The analysis warns that choice architecture policies in health insurance exchanges cannot be evaluated only by individual-level optimization. They must also be evaluated through market-level selection and repricing.
- **Methodological contribution.** The paper combines administrative claims data, predictive health risk modeling, a random coefficient plan choice model, and supply-side repricing into one welfare framework.

3 Data and measurement

3.1 Institutional setting and timeline

The paper studies employees of a large firm over a six-year window. The central institutional event is a major health insurance redesign. Before the redesign, employees were enrolled in an old plan environment. In year t_0 , the firm changed its menu and required active re-enrollment with no stated default option. In subsequent years, t_1 and t_2 , the PPO options remained the same but employees had an incumbent plan that continued as the default if they took no action.

Interpretation: This timing is the source of identification. The forced active choice year reveals preferences when inertia is minimized by design, while later years reveal whether consumers respond to new relative prices or remain attached to the status quo.

3.2 Plan menu

The main empirical analysis focuses on three PPO plans:

- PPO_{250} : low deductible and most comprehensive coverage;
- PPO_{500} : intermediate coverage;
- PPO_{1200} : high deductible plan with an HSA-linked feature.

The PPO plans have the same provider network and cover the same medical services. They differ mainly in premiums, deductibles, coinsurance, out-of-pocket maxima, and the HSA feature for PPO_{1200} .

Interpretation: Because the provider network is held fixed, the paper avoids a major confound common in health insurance choice: people may remain in a plan to keep their doctors. Here, inertia is not mainly network inertia.

3.3 Sample construction

The final structural estimation sample includes employees who are enrolled in a PPO option at t_0 , remain enrolled in a plan at the firm through at least t_1 , and have sufficient past medical data to predict ex ante health risk. Table 1 reports the step from all employees to those ever in PPO options and to the final estimation sample.

Interpretation: The sample is selected toward employees with stable employment and plan enrollment. This is useful for measuring inertia over time but means the estimates are most directly about continuing employees in this employer-sponsored setting.

3.4 Demographics and linked choices

The paper observes detailed demographics and linked choices, including:

- employee age, gender, income tier, family size, and job grouping;
- whether the employee is a manager or has a quantitative background;
- FSA enrollment, dental plan switching, and chronic condition indicators;
- plan choices over time and medical claims for employees and dependents.

Interpretation: FSA enrollment and dental switching are important because they proxy for active benefits engagement. Employees who actively manage other benefits should be less inertial in health plan choice.

3.5 Cost and health risk measurement

For each family k , plan j , and time t , the paper constructs an out-of-pocket expenditure distribution:

$$F_{kjt}(OOP).$$

The model uses detailed prior claims, diagnostic information, demographics, and medical prediction software to forecast future health risk. It separately models categories such as pharmacy, mental health, physician visits, and hospital/outpatient spending, then maps predicted medical spending into plan-specific out-of-pocket costs.

Interpretation: The key measurement problem is that insurance choice is made under uncertainty. The choice model therefore needs a distribution of possible costs, not just realized costs.

3.6 Inertia measures from descriptive evidence

The paper uses two descriptive tests before estimating the structural model.

- **New employee cohort test:** new employees at different dates face different current prices but have no incumbent plan, so their choices reveal active choice behavior.
- **Dominated plan test:** after the price change, PPO_{250} becomes dominated by PPO_{500} for some groups. Staying in PPO_{250} despite domination is direct evidence of inertia.

Interpretation: These tests are powerful because they do not require the full structural model. They show that inertia is visible directly in choice behavior.

3.7 Adverse selection measurement

Adverse selection is measured by comparing health costs of employees choosing different PPO options. Higher costs among enrollees in more comprehensive plans indicate that riskier employees select into more generous coverage.

Interpretation: This is the key reason reduced inertia can backfire. If healthier consumers leave generous coverage, the premium of that coverage rises, making selection more severe.

4 Models and empirical specifications

4.1 Expected utility from a health plan

For family k , plan j , and year t , expected utility is:

$$U_{kjt} = \int_0^\infty f_{kjt}(OOP) u_k(W_k, OOP, P_{kjt}, \mathbf{1}_{kj,t-1}) dOOP.$$

Here f_{kjt} is the density of out-of-pocket spending, P_{kjt} is the employee premium contribution, and $\mathbf{1}_{kj,t-1}$ indicates whether plan j was the previous plan.

Interpretation: The model treats health plan choice as a choice among risky financial lotteries. The family compares premiums plus uncertain out-of-pocket costs across plans.

4.2 CARA preferences

Families have CARA utility:

$$u_k(x) = -\frac{1}{\gamma_k} \exp(-\gamma_k x),$$

where γ_k is the family-specific coefficient of absolute risk aversion.

Interpretation: CARA preferences make risk aversion easy to estimate and let the model translate uncertain out-of-pocket costs into certainty equivalents. Higher γ_k means stronger taste for financial protection.

4.3 Consumption index and inertia

The consumption index for family k in plan j at time t is:

$$x = W_k - P_{kjt} - OOP + \eta(X_{kt}^B, Y_k)\mathbf{1}_{kj,t-1} + \delta_k(Y_k)\mathbf{1}_{j=PPO_{1200}} + \alpha H_{k,t-1}\mathbf{1}_{j=PPO_{250}} + \epsilon_{kjt}(Y_k).$$

- η is the monetary value of inertia or status quo attachment;
- δ_k captures horizontal differentiation for PPO_{1200} ;
- α captures an extra preference for PPO_{250} among very high-cost families;
- ϵ_{kjt} is an idiosyncratic preference shock.

Interpretation: Inertia enters as a utility bonus for staying with the incumbent plan. Equivalently, switching away requires the alternative plan to beat the incumbent by at least η dollars.

4.4 Heterogeneity in risk aversion and inertia

Risk aversion is modeled as:

$$\gamma_k(X_k^A) \sim N(\mu_\gamma(X_k^A), \sigma_\gamma^2), \quad \mu_\gamma(X_k^A) = \mu + \beta X_k^A.$$

Inertia is modeled as:

$$\eta(X_{kt}^B, Y_k) = \eta_0 + \eta_1 X_{kt}^B + \eta_2 Y_k.$$

Interpretation: Risk preferences are identified mainly from active choices across plans with different financial risk. Inertia is identified from persistence after premiums and plan values change.

4.5 Choice probability and estimation

The family chooses the plan with the highest latent utility:

$$Choice_{kt} = \arg \max_{j \in \mathcal{J}} U_{kjt}.$$

The model is estimated by simulated maximum likelihood over the sequence of choices from t_0 to t_2 .

Interpretation: The likelihood must be sequential because the current choice affects next year's incumbent plan, and therefore next year's inertia term.

4.6 Supply-side premium setting

In the counterfactuals, premiums reprice endogenously. Plan premiums are based on the average cost of enrollees in the prior period plus administrative costs, with a flat employer subsidy.

$$Premium_{j,t+1} \approx AverageCost_{j,t} + AdminFee_j - Subsidy_t.$$

Interpretation: This pricing rule transmits adverse selection into future premiums. If healthier families leave comprehensive coverage, the average cost of the remaining pool rises and the comprehensive plan becomes more expensive.

4.7 Counterfactual inertia reduction

The paper studies policies that reduce inertia from η to $Z\eta$, where:

$$Z \in [0, 1].$$

A smaller Z means a more effective intervention. For example, $Z = 0.25$ means inertia falls to one quarter of its estimated value.

Interpretation: The counterfactual is meant to represent policies such as clearer benefit descriptions, switching reminders, personalized information, or simplified choice architecture.

4.8 Welfare measurement

The welfare calculations compare certainty-equivalent utility under the baseline and counterfactual environments. The key distinction is between:

- **partial equilibrium welfare:** choices improve holding premiums fixed;
- **full equilibrium welfare:** choices change, risk pools change, and premiums reprice.

Interpretation: The sign reversal between partial and full equilibrium is the paper’s main policy result. Nudges can help people choose better at old prices but hurt once prices adjust to sharper sorting.

5 Identification and empirical design

5.1 Primary identification problem

The central empirical problem is to distinguish inertia from persistent unobserved preference heterogeneity. If a family remains in the same plan, this could mean either:

- the family has a stable preference for that plan; or
- the family is inertial and does not re-optimize after prices change.

Interpretation: Without observing active choices, persistence in plan enrollment cannot be interpreted as inertia. The paper’s institutional redesign solves this problem.

5.2 Forced active choice as identification

At t_0 , the firm redesigns the plan menu and forces employees to make an active choice with no stated default. In later years, the chosen plan becomes the default and employees can remain enrolled without taking action.

Active choice at $t_0 \Rightarrow$ identify risk preferences and plan tastes.

Passive default at $t_1, t_2 \Rightarrow$ identify inertia from non-switching.

Interpretation: The same individual is observed in both environments. This within-family design is much stronger than comparing different people with and without defaults.

5.3 Role of large premium changes

Inertia is identified because relative plan prices change meaningfully after t_0 . If prices did not change, persistence would not be very informative. Here, premium changes make some plans relatively much worse, and in some cases dominated.

Interpretation: Price changes create a revealed preference test. Families who actively chose a plan at old prices should switch when that plan becomes worse unless inertia or some omitted preference blocks them.

5.4 Why same networks matter

The PPO plans share providers and covered services. This matters because one common source of switching costs in health insurance is physician network attachment.

Interpretation: The estimated inertia is not simply about wanting to keep doctors. It is closer to a choice friction related to re-optimization, search, attention, or plan switching itself.

5.5 Identifying risk aversion versus plan differentiation

The model separately identifies:

- risk aversion γ_k from tradeoffs between PPO_{250} and PPO_{500} , which mainly differ in financial generosity;
- PPO_{1200} horizontal differentiation δ_k from choices involving the high-deductible HSA-linked option.

Interpretation: The three-plan structure gives more information than a binary menu. It helps separate risk protection preferences from HSA-specific preferences.

5.6 Identifying adverse selection

Adverse selection is identified from the relationship between plan choices and predicted or realized medical costs. If consumers in more comprehensive plans have higher expected costs, the market has selection.

Interpretation: The key result is not just that sick families buy more coverage. It is that this selection changes premiums when inertia falls and choices become more responsive.

5.7 Threats to identification

- **Private information.** Families may know health information not captured by the claims model.
- **Moral hazard.** Plan generosity might change medical spending rather than just out-of-pocket payments.
- **Myopia.** Consumers may make dynamic decisions if they anticipate future defaults or price changes.
- **Alternative inertia mechanisms.** Search costs, switching costs, inattention, and psychological default effects may have different welfare meanings.
- **Sample selection.** The final sample includes continuing employees with sufficient claims history.

Interpretation: The paper addresses these threats through rich claims data, robustness checks, and welfare treatments that vary whether inertia is counted as a real cost or only as a choice friction.

6 Tables and figures

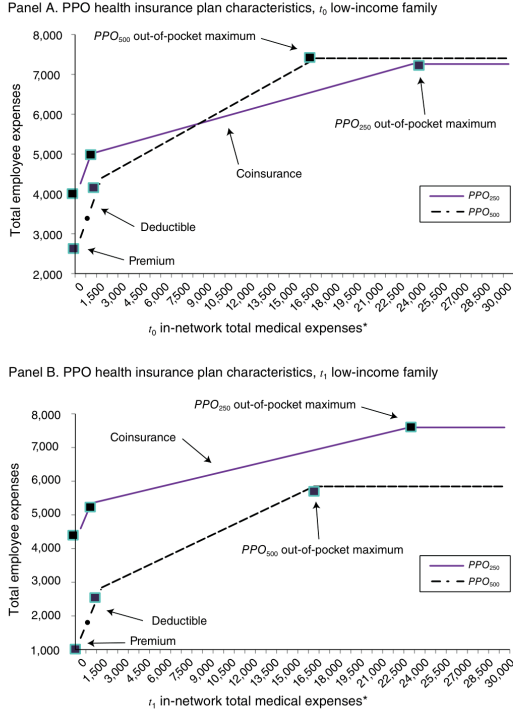
6.1 Figure 1: Financial Characteristics of PPO_{250} and PPO_{500} ; Figure 2: Evolution of Health Plan Premiums

Read: Figure 1 compares employee total expenses under PPO_{250} and PPO_{500} before and after the premium change. Figure 2 shows the large changes in employee premium contributions across income tiers and family status.

Economic message: The two figures explain why inertia is economically important. The price change is large enough that some prior choices become clearly unattractive, creating a strong test of whether consumers re-optimize.

Detailed interpretation: Panel B of Figure 1 is especially important: after the premium change, PPO_{500} can dominate PPO_{250} for low-income families. Thus, if families stay in PPO_{250} , the behavior is difficult to rationalize with standard static optimization. Figure 2 shows that the same type of plan can become much more or less expensive depending on income tier and family status, which creates rich variation for identifying plan value changes.

How it fits the paper: These figures motivate the whole empirical design. They show that the menu redesign was not cosmetic; it materially changed relative prices and therefore gives researchers a natural opportunity to detect inertia.



(a) **Figure 1: Financial Characteristics of PPO_{250} and PPO_{500} .** The original figure plots total employee expenses as a function of total in-network medical expenses under two PPO plans. *How to read it:* Compare where the purple and dashed black lines cross. A lower line is the cheaper plan for that spending level. *Why it matters:* The figure shows that price changes can make one plan strictly better than another, creating a direct test for inertia.

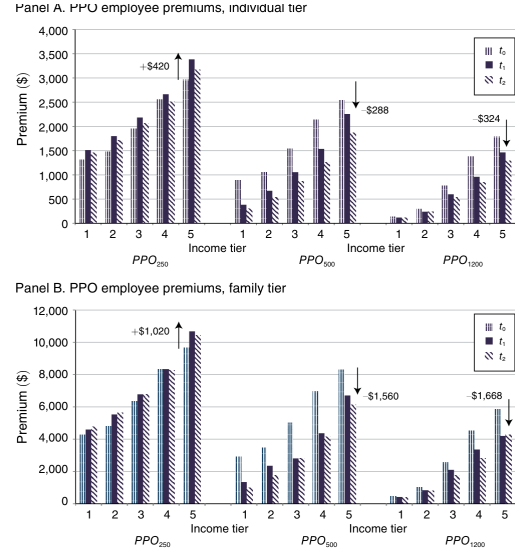


FIGURE 2. EVOLUTION OF HEALTH PLAN PREMIUMS

(b) **Figure 2: Evolution of Health Plan Premiums.** The original figure reports employee premium contributions for the PPO options across income tiers, family status, and years. *How to read it:* Bars compare the same plan-income tier over time. Large changes between t_0 and t_1 create incentives to switch. *Why it matters:* The figure documents the premium shock that drives much of the inertia test and the counterfactual repricing analysis.

6.2 Table 1: Descriptive Statistics; Table 2: New Employee Health Plan Choices

Read: Table 1 describes the sample construction and demographics. Table 2 compares plan choices of new employees across cohorts and years.

Economic message: Table 1 shows who is in the estimation sample; Table 2 provides a model-free test of inertia because new employees choose actively without an incumbent plan.

Detailed interpretation: Table 2 is a clean behavioral contrast. New entrants at t_1 choose PPO_{500} much more often than earlier cohorts do at the same date. Earlier cohorts remain more attached to choices made under old prices. Since the cohorts look similar demographically, the shift in new entrant choices points to active response to current prices rather than population changes.

How it fits the paper: Together, these tables establish the credibility of the sample and the need for an inertia term in the structural model.

TABLE 1—DESCRIPTIVE STATISTICS

Sample demographics	All employees	PPO ever	Final sample
<i>N</i> -Employee only	11,253	5,667	2,023
<i>N</i> -All family members	20,963	10,713	4,544
Mean employee age (median)	40.1 (37)	40.0 (37)	42.3 (44)
Gender (male) percent	46.7	46.3	46.7
<i>Income (percent)</i>			
Tier 1 (< \$41K)	33.9	31.9	19.0
Tier 2 (\$41K-\$72K)	39.5	39.7	40.5
Tier 3 (\$72K-\$124K)	17.9	18.6	25.0
Tier 4 (\$124K-\$176K)	5.2	5.4	7.8
Tier 5 (> \$176K)	3.5	4.4	7.7
<i>Family size (percent)</i>			
1	58.0	56.1	41.3
2	16.9	18.8	22.3
3	11.0	11.0	14.1
4+	14.1	14.1	22.3
<i>Staff grouping (percent)</i>			
Manager (percent)	23.2	25.1	37.5
White-collar (percent)	47.9	47.5	41.3
Blue-collar (percent)	28.9	27.3	21.1
<i>Additional demographics</i>			
Quantitative manager (percent)	12.8	13.3	20.7
Job tenure mean years (median)	7.2 (4)	7.1 (3)	10.1 (6)
Zip code population mean (median)	42,925 (42,005)	43,319 (42,005)	41,040 (40,175)
Zip code income mean (median)	\$56,070 (\$55,659)	\$56,322 (\$55,659)	\$60,948 (\$57,393)
Zip code house value mean (median)	\$226,886 (\$204,500)	\$230,083 (\$209,400)	\$245,380 (\$213,300)

Notes: This table presents summary demographic statistics for the population we study. The first column describes demographics for the entire sample, whether or not they ever enroll in insurance with the firm. The second column summarizes these variables for the sample of individuals who ever enroll in a PPO option, the choices we focus on in the empirical analysis. The third column describes our final estimation sample, which includes those employees who (i) are enrolled in PPO₋₁ at t_{-1} and (ii) remain enrolled in any plan at the firm through at least t_1 . Comparing the

(a) **Table 1: Descriptive Statistics.** The original table reports demographics for all employees, PPO enrollees, and the final sample.

How to read it: Compare the final sample column to the broader employee columns to see selection into the estimation sample. *Why it matters:* The final sample is somewhat older, higher income, and more family-oriented, which matters for interpreting inertia and welfare.

TABLE 2—NEW EMPLOYEE HEALTH PLAN CHOICES

New enrollee analysis	New enrollee t_{-1}	New enrollee t_0	New enrollee t_1
<i>N, t₀</i>	1,056	1,377	—
<i>N, t₁</i>	784	1,267	1,305
<i>t₀ Choices</i>			
PPO ₂₅₀	259 (25%)	287 (21%)	—
PPO ₅₀₀	205 (19%)	306 (23%)	—
PPO ₁₂₀₀	155 (15%)	236 (17%)	—
HMO ₁	238 (23%)	278 (20%)	—
HMO ₂	199 (18%)	270 (19%)	—
<i>t₁ Choices</i>			
PPO ₂₅₀	182 (23%)	253 (20%)	142 (11%)
PPO ₅₀₀	201 (26%)	324 (26%)	562 (43%)
PPO ₁₂₀₀	95 (12%)	194 (15%)	188 (14%)
HMO ₁	171 (22%)	257 (20%)	262 (20%)
HMO ₂	135 (17%)	239 (19%)	151 (12%)
<i>Demographics</i>			
Mean age	33	33	32
Median age	31	31	31
Female percent	56%	54%	53%
Manager percent	20%	18%	19%
FSA enroll percent	15%	12%	14%
Dental enroll percent	88%	86%	86%
Median (mean) expense t_1	844 (4,758)	899 (5,723)	—
Income tier 1	48%	50%	47%
Income tier 2	33%	31%	32%
Income tier 3	10%	10%	12%
Income tier 4	5%	4%	4%
Income tier 5	4%	5%	5%

Notes: This table describes the choice behavior of new employees at the firm over several consecutive years and presents our first model-free test of inertia. Each column describes one cohort of new employees at the firm, corresponding to a specific year of arrival. First, the chart describes the health insurance choices made by these cohorts in year t_0 (the year of the insurance plan menu change) and in the following year, t_1 . The last part of the chart lists the demographics for each cohort of new arrivals at the time of their arrival. Given the very similar demographic profiles and large sample size for each cohort, if there is no inertia, the t_1 choices of employees who

(b) **Table 2: New Employee Health Plan Choices.** The original table reports plan choices for new employee cohorts at different dates.

How to read it: Compare t_1 choices of old cohorts with t_1 choices of new entrants.

Why it matters: New entrants respond more strongly to current premiums, while incumbents exhibit persistence, consistent with inertia.

6.3 Table 3: Dominated Plan Choice Analysis; Table 4: Adverse Selection and Employee Costs

Read: Table 3 studies employees whose plan becomes dominated and compares stayers to switchers. Table 4 studies whether higher-risk families select into more comprehensive plans.

Economic message: These tables provide the two descriptive ingredients of the paper: inertia and adverse selection. Table 3 shows non-optimization; Table 4 shows risk-based sorting.

Detailed interpretation: Table 3 shows that many families remain in PPO₂₅₀ even after it becomes dominated by PPO₅₀₀, losing hundreds of dollars. Switchers look more active in other benefits decisions, such as FSA enrollment and dental switching. Table 4 shows that PPO₂₅₀ enrollees have higher past and realized costs than less comprehensive plan enrollees, meaning that the generous plan contains a sicker risk pool.

How it fits the paper: The later counterfactuals depend on both forces. If there were inertia without adverse selection, nudges would mostly help. If there were adverse selection without inertia, there would be no behavioral friction to remove. The interaction of the two creates the welfare reversal.

TABLE 3—DOMINATED PLAN CHOICE ANALYSIS

Dominated plan analysis	t_1 Dominated stay	t_1 Dominated switch	t_2 Dominated stay	t_2 Dominated switch
N	498	61	378	126
Minimum money lost ^a	\$374	\$453	\$396	\$306
PPO_{100}	—	44 (72%)	—	103 (81%)
PPO_{120}	—	4 (7%)	—	6 (5%)
Any HMO	—	13 (21%)	—	17 (14%)
FSA t_1	25.4%	32.1%	27.2%	28.6%
FSA t_2	—	—	28.1%	30.9%
Dental switch t_1	4.3%	14.1%	3.5%	10.9%
Dental switch t_2	—	—	6.9%	17.2%
Age (mean)	44.9	38.3	46.2	41.4
Income tier (mean) ^b	1.6	1.4	1.6	1.7
Quant. manager	11%	8%	11%	11%
Single (percent)	40%	41%	40%	33%
Male (percent)	42%	46%	39%	55%
All plan analysis	PPO_{250} stay t_1	PPO_{250} switch t_1	All plans t_1 stay	All plans t_1 switch
Sample size	1,626	174	2,786	384
FSA t_1 enrollee	31%	41%	25%	39%
Dental switch	3.2%	13.1%	3.8%	14.5%
Age (mean)	48.3	40.6	44.0	39.1
Income tier (mean) ^b	2.5	2.2	2.3	2.1
Quant. manager	20%	17%	17%	14%
Single (percent)	50%	56%	53%	59%
Male (percent)	48%	42%	49%	40%

Notes: This top panel in this table profiles the choices and demographics of the employees enrolled in PPO_{250} at t_0 who (i) continue to enroll in a firm plan in t_1 and (ii) have PPO_{250} become dominated for them at t_1 . The majority of these employees (498 out of 559 (89 percent)) remain in PPO_{250} even after it becomes dominated by PPO_{500} with 378 of 504 (25 percent) still remaining in this plan at t_2 . People who do switch are more likely to exhibit a pattern of active choice behavior in general as evidenced by their higher FSA enrollments and level of dental plan switching. Apart from this, these populations are similar though switchers in this group are slightly younger. The bottom panel studies the profiles of those who switch at t_1 and those who don't for the two groups of (i) PPO_{250} enrollees at t_0 and (ii) the entire universe of PPO plan enrollees present in t_0 and t_1 . This reveals a similar pattern of active decision making as switchers in these populations are also more likely to enroll in FSAs and switch dental plans.

^aMean of lower bound on money lost from enrolling in PPO_{250} instead of PPO_{500} .

^bIncome tier has values 1–5 where 1 is lowest income and 5 is highest. Tiers 1–4 are approximately \$40,000 increments, starting at 0, while tier 5 is everything greater than approximately \$175,000.

(a) **Table 3: Dominated Plan Choice Analysis.** The original table profiles employees facing dominated plan choices and compares stayers to switchers.
How to read it: Focus on the high share of dominated stayers and on activity proxies such as FSA enrollment and dental switching.
Why it matters: The table gives direct, non-structural evidence that many employees fail to switch even when a plan is strictly worse.

TABLE 4—ADVERSE SELECTION AND EMPLOYEE COSTS

Final sample total expenses	PPO_{100}	PPO_{250}	PPO_{500}	PPO_{1200}
<i>Family t_{-1} total expenses (\$)</i>				
t_{-1}				
N employees (mean family size)	2,022 (2.24)	—	—	—
Mean (median)	13,331 (4,916)	—	—	—
25th percentile	1,257	—	—	—
75th percentile	13,022	—	—	—
t_0				
N (mean family size)	—	1,328 (2.18)	414 (2.20)	280 (2.53)
Mean (median)	—	16,976 (6,628)	6,151 (2,244)	6,742 (2,958)
25th percentile	—	2,041	554	658
75th percentile	—	16,135	6,989	8,073
t_1				
N (mean family size)	—	1,244 (2.19)	546 (2.19)	232 (2.57)
Mean (median)	—	17,270 (6,651)	7,759 (2,659)	6,008 (2,815)
25th percentile	—	2,041	708	589
75th percentile	—	16,707	8,588	7,191
<i>Individual category expenses (dollars)</i>				
Pharmacy				
Mean	973	1,420	586	388
Median	81	246	72	22
Mental health (> 0)				
Mean	2,401	2,228	1,744	2,134
Median	1,260	1,211	1,243	924
Hospital/physician				
Mean	4,588	5,772	2,537	2,722
Median	428	717	255	366
Physician OV				
Mean	461	571	381	223
Median	278	356	226	120

Notes: This table investigates the extent of adverse selection across PPO options after the t_0 menu change for those in the final estimation sample. All individuals in this sample were enrolled in PPO_{100} in t_{-1} and continue to be enrolled in some plan at the firm for the following two years. The numbers in the table for all choices represent t_{-1} total claims in dollars so that these costs can proxy for health risk without being confounded by moral hazard (t_0 and t_1 cost differences could be the result of selection or moral hazard). The table reveals that those who choose PPO_{250} have much higher expenditures at t_{-1} than those who choose the other two plans, implying substantial selection on observables in the vein of Finkelstein and Poterba (2006). The bottom panel presents a breakdown of these costs

(b) **Table 4: Adverse Selection and Employee Costs.** The original table reports medical costs by PPO plan choice and expenditure category.
How to read it: Compare costs of PPO_{250} enrollees to PPO_{500} and PPO_{1200} enrollees.
Why it matters: More comprehensive coverage attracts higher-cost families, so plan prices are sensitive to enrollment composition.

6.4 Table 5: Choice Model Parameter Estimates

Read: Table 5 reports the estimated structural parameters for the primary specification and robustness specifications.

Economic message: The large inertia parameters turn the descriptive persistence into a dollar value. The model implies that families are willing to forgo around two thousand dollars to remain with the status quo plan.

Detailed interpretation: The inertia intercept is large for singles and even larger for families. FSA enrollment reduces inertia, consistent with more active benefits management. Risk aversion estimates imply that incremental insurance has value, so choosing more comprehensive coverage is not automatically a mistake. The table also shows that the main estimates are robust to alternative plan sets, moral hazard assumptions, risk distribution assumptions, and preference shock specifications.

How it fits the paper: Table 5 is the bridge between descriptive evidence and counterfactual welfare. Without structural estimates of inertia and risk preferences, the paper could not simulate how people would choose under counterfactual policies.

TABLE 5—CHOICE MODEL PARAMETER ESTIMATES

Empirical model results Parameter	Primary	Two plan	MH robust	γ Robust	ϵ Robust
Inertia—single, η_0	1,729 (28)	1,686 (82)	1,859 (107)	2,430 (116)	1,944 (150)
Inertia—family, $\eta_0 + \eta_2$	2,480 (26)	2,401 (73)	2,355 (113)	3,006 (94)	2,365 (34)
Inertia—FSA enroll, η_1	−551 (56)	−355 (78)	−669 (155)	−723 (131)	−417 (50)
Inertia—income, η_1	−32 (13)	−130 (22)	−59 (15)	−8 (43)	−7 (15)
Inertia—quantitative, η_1	5 (138)	−122 (110)	−40 (80)	−537 (223)	−6 (92)
Inertia—manager, η_1	198 (292)	464 (106)	277 (164)	875 (200)	224 (244)
Inertia—chronic condition, η_1	80 (46)	26 (72)	29 (67)	−221 (148)	67 (35)
Inertia—salient change, η_1	156 (83)	13 (102)	95 (60)	61 (212)	123 (54)
Inertia— PPO_{1200} , η_1	−19 (184)	—	−32 (46)	−327 (122)	−113 (52)
Inertia—total pop. mean, η [pop. standard deviation]	2,032 [446]	1,802 [416]	1,886 [387]	1,914 [731]	1,986 [316]
Risk aversion mean— intercept, μ_γ	2.32×10^{-4} (9.0×10^{-6})	3.25×10^{-4} (2.2×10^{-5})	2.31×10^{-4} (1.1×10^{-5})	−8.94 (0.43)	1.90×10^{-4} (1.0×10^{-5})
Risk aversion mean—income, β	2.90×10^{-5} (4.0×10^{-6})	6.11×10^{-5} (9.0×10^{-6})	1.80×10^{-5} (3.0×10^{-6})	0.07 (0.016)	2.40×10^{-5} (3.0×10^{-6})
Risk aversion mean—age, β	2.27×10^{-6} (1.7×10^{-7})	7.16×10^{-6} (4.6×10^{-7})	3.45×10^{-6} (1.8×10^{-7})	0.28 (0.011)	2.59×10^{-6} (1.5×10^{-7})
Risk aversion standard deviation, σ_γ	1.88×10^{-4} (6.6×10^{-5})	4.06×10^{-4} (2.3×10^{-5})	1.27×10^{-4} (6.0×10^{-6})	1.37 (0.06)	1.04×10^{-4} (5.9×10^{-5})
CDHP—single—RC mean, δ	−2,912 (754)	—	−2,801 (416)	−2,985 (85)	−2,833 (130)
CDHP—single—RC SD, σ_δ	843 (431)	—	1,070 (139)	989 (70)	1,141 (113)
CDHP—family—RC mean, δ	−2,871 (73)	—	−2,614 (115)	−5,344 (134)	−2,932 (40)
CDHP—family—RC SD, σ_δ	897 (28)	—	1,149 (132)	2,179 (80)	1,013 (31)
High total cost— PPO_{250} , α	856 (50)	763 (55)	607 (55)	1,386 (264)	860 (66)
ϵ_{500} , σ —single	204 (13)	57 (25)	51 (30)	50 (55)	—
ϵ_{1200} , σ —single	502 (475)	—	647 (228)	161 (72)	—
ϵ_{500} , σ —family	329 (25)	590 (68)	789 (28)	90 (89)	—
ϵ_{1200} , σ —family	811 (25)	—	715 (44)	676 (426)	—

Table 5: Choice Model Parameter Estimates. The original table presents the estimated inertia, risk-aversion, plan-differentiation, and preference-shock parameters.

How to read it: Read the first rows as dollar values of inertia and the middle rows as heterogeneity in risk aversion and PPO_{1200} differentiation.

Why it matters: The estimated inertia is large and stable across robustness checks, giving the counterfactual exercise empirical bite.

6.5 Figure 3: Impact of Reduced Inertia on Choices and Costs; Table 6: Welfare Impact of Reduced Inertia: η to 0.25η

Read: Figure 3 shows enrollment and average cost paths under baseline inertia and reduced inertia. Table 6 quantifies welfare effects for a policy that reduces inertia to one quarter of its baseline value.

Economic message: These visuals are the paper’s central “nudging hurts” evidence. Reducing inertia shifts healthier enrollees away from the comprehensive plan, increasing its average cost and premium.

Detailed interpretation: Figure 3 shows that reduced inertia moves market share away from PPO_{250} and toward PPO_{500} . Average costs in PPO_{250} rise relative to the baseline, which is the adverse selection mechanism. Table 6 reveals the distributional consequences: switchers gain, but non-switchers lose because premiums reprice against them. The population average effect is negative even though individual optimization conditional on old prices improves.

How it fits the paper: The figure and table make the distinction between partial equilibrium and full equilibrium visible. The policy helps people choose better at existing prices, but those choices change the prices.

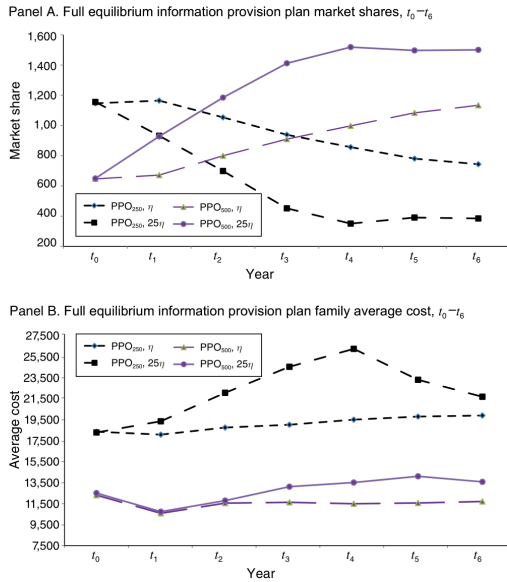


FIGURE 3. IMPACT OF REDUCED INERTIA ON CHOICES AND COSTS: WHEN NUDGING HURTS

Notes: Panel A presents the time path of choices for PPO_{250} and PPO_{500} with and without the

(a) **Figure 3: Impact of Reduced Inertia on Choices and Costs: When Nudging Hurts.** The original figure compares plan market shares and average costs under full inertia and reduced inertia.
How to read it: Look at the divergence between the baseline and reduced-inertia paths for PPO_{250} and PPO_{500} .
Why it matters: Reduced inertia intensifies sorting and raises the cost of the comprehensive plan.

TABLE 6—WELFARE IMPACT OF REDUCED INERTIA: η TO 0.25 η

Plan re-pricing welfare analysis	t_1	t_2	t_4	t_6	Avg. t_1 – t_6
<i>Mean Δ TS</i>					
Population	–\$63	–\$104	–\$144	–\$118	–\$115
Switcher population percent	51	49	48	53	49
Switchers only	\$86	\$175	\$245	\$242	\$186
Non-switchers only	–\$205	–\$391	–\$555	–\$432	–\$442
High expense population percent	10	11	11	11	11
High expense	\$26	\$106	\$119	\$65	\$62
Non-high expense	–\$73	–\$130	–\$177	–\$141	–\$137
Single population percent	47	46	46	46	46
Single	–\$249	–\$367	–\$414	–\$195	–\$319
W/dependents	\$99	\$124	\$89	–\$51	\$61
Low income population percent	40	41	41	41	41
Low income	–\$81	–\$218	–\$282	–\$178	–\$200
High income	–\$36	\$62	\$57	–\$30	\$0
<i>Welfare change: percent premiums</i>					
Mean employee premium	\$1,471	\$1,591	\$1,455	\$1,259	\$1,500
Welfare change population	–4.8	–6.5	–9.9	–9.4	–7.7
Welfare change switchers	5.6	11.0	16.9	19.2	12.4
Welfare change non-switchers	–13.9	–24.6	–38.1	–34.3	–29.4
<i>Welfare change: percent total spending</i>					
Mean total employee spending	\$3,755	\$4,097	\$4,022	\$3,862	\$4,015
Welfare change population	–1.7	–2.5	–3.6	–3.06	–2.9
Welfare change switchers	2.3	4.3	6.1	6.3	4.6
Welfare change non-switchers	–5.5	–9.5	–13.8	–11.2	–11.0
<i>Welfare change: percent $\parallel$ CEQ $\parallel$ Loss</i>					
Mean total $\parallel$ CEQ $\parallel$ Loss	\$5,888	\$6,264	\$6,207	\$6,065	\$6,190
Welfare change population	–1.1	–1.7	–2.3	–2.0	–1.9
Welfare change switchers	1.5	2.8	4.0	4.0	3.0
Welfare change non-switchers	–3.5	–6.2	–8.9	–7.1	–7.1

Notes: This table presents the welfare results of the endogenous insurance pricing policy counterfactual for the case where inertia is reduced from η to 0.25 η . We present the change in the mean per employee per year certainty equivalent moving from the simulation with full inertia to the one with reduced inertia. In addition to studying the effect of the policy on efficiency, we study the distributional effects based on four categorizations (i) “switchers”, or people who are in a different plan at time t under the policy intervention than without it; (ii) an indicator of whether or not the family has high health costs relative to its coverage tier; (iii) whether an employee is single or covers dependents; and (iv) whether an employee has high or low income. The three welfare metrics we present take the ratio of the change in certainty equivalent with respect to (i) total employee premiums, (ii) total employee spending, and (iii) the absolute value of the certainty equivalent loss.

(b) **Table 6: Welfare Impact of Reduced Inertia: η to 0.25 η .** The original table reports welfare changes overall and by subgroup when inertia is reduced by 75 percent.
How to read it: Compare switchers and non-switchers. Switchers gain from better choices; non-switchers lose from repricing.
Why it matters: Aggregate welfare falls because the premium externality from adverse selection dominates the private choice improvement.

6.6 Table 7: Welfare Impact of Reduced Inertia: Differential Intervention Effectiveness; Table 8: Different Welfare Treatments of Inertia

Read: Table 7 varies the effectiveness of inertia reduction. Table 8 varies whether inertia itself is counted as a welfare-relevant cost.

Economic message: The negative welfare effect is not a knife-edge result. It appears over a broad range of policies and welfare treatments, although magnitudes differ.

Detailed interpretation: Table 7 shows that more effective reductions in inertia often produce larger adverse selection losses. The first-best column shows that eliminating adverse selection would be valuable, but the realistic nudge affects choices without fixing risk pricing. Table 8 asks whether overcoming inertia is itself a welfare benefit. Even when part of inertia is treated as a real cost, many interventions still generate negative welfare because repricing hurts many consumers.

How it fits the paper: These tables are important for policy interpretation. They show that the conclusion is not simply an artifact of assuming inertia is pure inattention. The interaction with adverse selection remains central under alternative welfare treatments.

TABLE 7—WELFARE IMPACT OF REDUCED INERTIA: DIFFERENTIAL INTERVENTION EFFECTIVENESS

Endogenous plan re-pricing welfare analysis reduction in inertia	First-best	Baseline	0.75 η	0.5 η	0.25 η	0
<i>Mean Δ TS (percent of premiums)</i>						
Population	\$123 (8.2)	— (—)	−\$41 (−2.7)	−\$73 (−4.9)	−\$115 (−7.7)	−\$107 (−7.1)
Switchers	−\$538 (−35.9)	— (—)	\$1,017 (67.8)	\$766 (51.0)	\$186 (12.4)	\$118 (7.9)
Non-switchers	\$953 (63.5)	— (—)	−\$249 (−16.6)	−\$371 (−24.8)	−\$442 (−29.4)	−\$382 (−25.4)
High expense	\$936 (62.4)	— (—)	\$38 (2.6)	\$84 (5.6)	\$62 (4.2)	\$121 (8.1)
Non-high expense	\$22 (1.5)	— (—)	−\$52 (−3.5)	−\$93 (−6.2)	−\$137 (−9.2)	−\$136 (−9.1)
Single	−\$683 (−45.5)	— (—)	−\$153 (−10.2)	−\$295 (−19.7)	−\$319 (−21.2)	−\$286 (−19.0)
Family	\$826 (55)	— (—)	−\$54 (3.6)	\$119 (7.9)	\$61 (4.1)	\$47 (3.1)
Low income	−\$349 (−23.3)	— (—)	−\$75 (−5.0)	−\$153 (−10.2)	−\$200 (−13.3)	−\$190 (−12.7)
High income	\$806 (53.7)	— (—)	\$10 (0.6)	\$43 (2.9)	\$0 (0)	\$13 (0.9)

Notes: This table shows the welfare change of a range of policy interventions, in terms of effectiveness, relative to the baseline where preferences are as estimated in Table 5. In addition, we present results on the welfare loss from adverse selection in the actual environment relative to the first-best. The chart reports the change in the mean per employee per year certainty equivalent in each environment, relative to the baseline case. In parentheses, we include the percentage corresponding to this certainty equivalent change divided by mean employee premiums paid per employee per year. Column 1 shows how the first-best compares to the baseline and reveals that the mean welfare loss from adverse selection in the current information environment is \$123 or 8.2 percent of total premiums paid in the baseline. Columns 3 through 6 correspond to different counterfactual environments where inertia has been reduced relative to the baseline. We study four cases, when inertia is assumed to be 75 percent, 50 percent, 25 percent, and 0 percent of baseline inertia respectively. We report welfare results for the population as well as different segments of the population. The 95 percent counterfactual is examined in more detail in Table 6.

(a) **Table 7: Welfare Impact of Reduced Inertia: Differential Intervention Effectiveness.** The original table compares policies with different degrees of inertia reduction. *How to read it:* Read columns from baseline toward zero inertia; lower inertia generally induces more switching and stronger selection effects. *Why it matters:* More effective nudges can produce larger equilibrium losses when they sharpen adverse selection.

TABLE 8—DIFFERENT WELFARE TREATMENTS OF INERTIA

Endogenous plan re-pricing Welfare treatment of inertia	η	0.75 η	0.5 η	0.25 η	0
Avg. t_1 - t_0					
Inertia cost / switcher	1,963	1,489	988	493	0
Switcher %	9	13	14	17	20
Avg. inertia pop.	185	188	142	83	0
Welfare impact	η	0.75 η	0.5 η	0.25 η	0
$\kappa = 0$	Welfare relevant inertia	0	0	0	0
	Δ TS (% premiums)	—	−\$41 (−2.7)	−\$73 (−4.9)	−\$115 (−7.7)
$\kappa = 0.25$	Welfare relevant inertia	46	47	36	21
	Δ TS (% premiums)	—	−\$42 (−2.8)	−\$63 (−4.2)	−\$90 (−6.0)
$\kappa = 0.5$	Welfare relevant inertia	93	94	71	42
	Δ TS (% premiums)	—	−\$42 (−2.8)	−\$51 (−3.4)	−\$64 (−4.3)
$\kappa = 1$	Welfare relevant inertia	185	188	142	83
	Δ TS (% premiums)	—	−\$44 (−2.9)	−\$30 (−2.0)	−\$13 (−0.9)

Notes: Table 8 expands the welfare analysis to account for the possibility that some proportion of estimated, and subsequently reduced, inertia should be included in the welfare analysis. Tables 6 and 7 present results conditional on $\kappa = 0$ (overcoming inertia is not welfare relevant cost) while this table presents results across the range of κ from 0 to 1 (overcoming inertia is purely a direct and welfare relevant cost). The top panel of this table studies the profile of maximum incurred tangible costs of inertia for different Z from t_1 to t_0 , while the bottom panel assesses the welfare impact of these interventions as a function of κ . The table reveals that, for almost all combinations of κ and Z , there is a negative welfare impact from reduced inertia and better consumer decisions.

(b) **Table 8: Different Welfare Treatments of Inertia.** The original table varies the share of inertia treated as welfare relevant and reports welfare effects across policy strengths. *How to read it:* Rows vary κ , the share of inertia counted as a real welfare cost; columns vary the residual inertia level. *Why it matters:* Even when reducing inertia has direct welfare value, adverse selection can still make many nudges harmful in this market.

7 Short summary

- The paper studies how inertia interacts with adverse selection in employer health insurance.
- A forced active choice event identifies inertia separately from persistent preference heterogeneity.
- Descriptive evidence shows that many employees remain in dominated plans after premium changes.
- Cost evidence shows adverse selection: higher-risk families select into more comprehensive plans.
- A structural model estimates risk preferences, inertia, plan differentiation, and ex ante health risk.
- The average value of inertia is about \$2,032 per family per year in the primary model.
- Counterfactual policies reduce inertia by scaling η down to $Z\eta$.
- Holding prices fixed, reducing inertia improves individual choices.
- With endogenous repricing, reducing inertia worsens selection and lowers average welfare in this setting.
- Switchers gain, but non-switchers lose from higher premiums and market re-sorting.
- The paper’s main policy lesson is that nudges in insurance markets must account for equilibrium price effects.

8 Conclusion

8.1 Core conclusion

Handel shows that reducing inertia in health insurance markets can lower welfare when adverse selection is present. The mechanism is not that consumers become worse at choosing. Rather, better individual choices change the composition of insurance pools, causing premiums to reprice and making comprehensive coverage

less attractive for relatively healthy consumers. The same nudge that helps a person optimize can harm the market equilibrium.

8.2 Economic intuition

Inertia keeps some healthy people in relatively generous plans. This softens adverse selection because the generous plan contains a more balanced risk pool. When inertia is reduced, healthier consumers move to cheaper and less comprehensive plans. The comprehensive plan becomes sicker on average, its premium rises, and still more healthy consumers leave. The welfare loss comes from this equilibrium feedback, not from the first-order effect of making people more attentive.

8.3 Policy implications

The paper does not argue that consumer inertia is good in general. Instead, it shows that policy design must consider multiple frictions jointly. Information provision, simplified plan labels, active choice mandates, and switching reminders may improve individual choices, but in health insurance they also change selection. Effective policy may require combining nudges with risk adjustment, subsidies, plan standardization, or pricing rules that blunt selection incentives.

8.4 Interpretation of the welfare result

The negative welfare effect is specific to the studied population and plan environment. The sign could differ if the population had different risk preferences, if the plan menu were different, if subsidies were structured differently, or if risk adjustment were stronger. The generalizable lesson is not that inertia should be preserved. The generalizable lesson is that behavioral frictions and adverse selection interact, so removing one distortion can amplify another.

8.5 Limitations and open questions

The paper models inertia as a monetary switching cost but does not distinguish among inattention, hassle costs, search costs, psychological default effects, or rational switching costs. These mechanisms have different normative meanings. The cost model assumes limited private information and limited moral hazard, though the paper argues these are unlikely to overturn the main results. Finally, the setting is one large employer, so external validity depends on whether exchanges and employer markets have similar sorting and repricing dynamics.

8.6 Takeaway

The enduring contribution is to shift the evaluation of choice architecture from individual optimization to market equilibrium. In insurance markets, nudges do not just move people; they move prices. When prices depend on who enrolls, behavioral policy must be evaluated together with adverse selection.